

A Convex-Analytic Formulation of the Møller–Plesset Adiabatic Connection

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Density-Functional Theory

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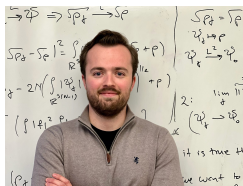
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Outline

- **Density-Functional Theory Adiabatic Connection**
- **Møller–Plesset Adiabatic Connection (MPAC)**
 - Local Adiabatic Connection
 - Hellmann-Feynman MPAC
- **Convex Analytic AC Formalism**
 - Integral AC Energy
 - Properties of the AC Integrand
- **A Virial Relation**
 - Exchange Only Virial Relation
- **Conclusion and Outlook**

Density-Functional Theory (DFT)

DFT Hamiltonian: $\hat{H}_{\text{DFT}} = \hat{T} + \hat{V}_{\text{ee}} + \hat{V}$ acting on $\mathcal{H}_N = \bigwedge_{j=1}^N \mathcal{H}_1$ for $v \in \mathcal{V} = L^{3/2} + L^\infty$, where

$$\hat{T} = -\frac{1}{2} \sum_{j=1}^N \nabla_{\mathbf{r}_j}^2, \quad \hat{V}_{\text{ee}} = \frac{1}{2} \sum_{j=1}^N \sum_{k \neq j}^N \frac{1}{|\mathbf{r}_j - \mathbf{r}_k|}, \quad \hat{V} = \sum_{j=1}^N v(\mathbf{r}_j).$$

Set of admissible densities: $\mathcal{P}_N = \{\rho^{1/2} \in H^1(\mathbb{R}^3), \rho \geq 0, \int \rho = N\}$, where

$$\Psi \rightarrow \rho : \rho(\mathbf{r}) = N \int \cdots \int |\Psi(\mathbf{r}, \mathbf{r}_2, \dots, \mathbf{r}_N)|^2 d\mathbf{r}_2 \cdots d\mathbf{r}_N \text{ for } \Psi \in \mathcal{H}_N.$$

Ground-state energy:

$$E_0(v) = \inf_{\rho \in \mathcal{P}_N} E_{\text{DFT}}[\rho] = \inf_{\rho \in \mathcal{P}_N} \left\{ F[\rho] + \int v \rho \right\} \text{ with } F[\rho] = \inf_{\|\Psi\|=1, \Psi \rightarrow \rho} \langle \Psi | \hat{T} + \hat{V}_{\text{ee}} | \Psi \rangle.$$

Let ρ be the ground-state density for E_{DFT} , $\Psi_{\text{DFT}}[\rho]$ the minimizer for $F[\rho]$ and

$$\Phi_{\text{DFT}}[\rho] = \arg \min_{\|\Phi\|=1, \Phi_{\text{DFT}} \rightarrow \rho} \langle \Phi | \hat{T} | \Phi \rangle,$$

DFT Correlation Energy

$$E_{\text{c,DFT}}[\rho] = \langle \Psi_{\text{DFT}}[\rho] | \hat{H} | \Psi_{\text{DFT}}[\rho] \rangle - \langle \Phi_{\text{DFT}}[\rho] | \hat{H} | \Phi_{\text{DFT}}[\rho] \rangle$$

DFT Adiabatic Connection (AC)

DFT AC Hamiltonian: $\hat{H}_{\text{DFT}}^\lambda = \hat{T} + \lambda \hat{V}_{\text{ee}} + \hat{V}^\lambda$ for $\lambda \in \mathbb{R}_0^+$ and $v^\lambda \in \mathcal{V}$ such that $\Psi_{\text{DFT}}^\lambda \rightarrow \rho$.

DFT AC Energy: $E_{\text{DFT}}^\lambda[\rho] = F^\lambda[\rho] + \int v^\lambda \rho$ with $F^\lambda[\rho] = \langle \Psi_{\text{DFT}}^\lambda[\rho] | \hat{T} + \lambda \hat{V}_{\text{ee}} | \Psi_{\text{DFT}}^\lambda[\rho] \rangle$.

Physical system $\lambda = 1$

$$E_{\text{DFT}}^{\lambda=1}[\rho] = E_{\text{DFT}}[\rho] \text{ and } \Psi_{\text{DFT}}^{\lambda=1}[\rho] = \Psi_{\text{DFT}}[\rho]$$

Non-interacting system $\lambda = 0$

$$E_{\text{DFT}}^{\lambda=0}[\rho] = \langle \Psi_{\text{DFT}}^{\lambda=0}[\rho] | \hat{T} | \Psi_{\text{DFT}}^{\lambda=0}[\rho] \rangle + \int v^{\lambda=0} \rho$$

and $\Psi_{\text{DFT}}^{\lambda=0}[\rho] = \Phi_{\text{DFT}}[\rho]$.

DFT AC Correlation Energy

$$E_{\text{c,DFT}}[\rho] = \langle \Psi_{\text{DFT}}^{\lambda=1}[\rho] | \hat{H} | \Psi_{\text{DFT}}^{\lambda=1}[\rho] \rangle - \langle \Psi_{\text{DFT}}^{\lambda=0}[\rho] | \hat{H} | \Psi_{\text{DFT}}^{\lambda=0}[\rho] \rangle = \int_0^1 \mathcal{E}_{\text{c,DFT}}^\lambda[\rho] d\lambda$$

DFT AC integrand:

$$\mathcal{E}_{\text{c,DFT}}^\lambda[\rho] = \langle \Psi_{\text{DFT}}^\lambda[\rho] | \hat{V}_{\text{ee}} | \Psi_{\text{DFT}}^\lambda[\rho] \rangle - \langle \Phi_{\text{DFT}}[\rho] | \hat{V}_{\text{ee}} | \Phi_{\text{DFT}}[\rho] \rangle.$$

Møller–Plesset Adiabatic Connection (MPAC)

MPAC Hamiltonian: $\hat{H}_{\text{MPAC}}^\lambda = \hat{T} + \hat{V} + \lambda \hat{V}_{\text{ee}} + (1 - \lambda)(\hat{J} + \hat{K})$.

Consider a non-degenerate Hartree-Fock (HF) state Φ_{HF} and density $\Phi_{\text{HF}} \rightarrow \rho_{\text{HF}}$,

- $\hat{J}$: Coulomb operator with

$$E_{\text{H}}[\rho_{\text{HF}}] = \frac{1}{2} \langle \Phi_{\text{HF}} | \hat{J} | \Phi_{\text{HF}} \rangle = \frac{1}{2} \int \int \frac{\rho_{\text{HF}}(\mathbf{r}) \rho_{\text{HF}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'.$$

- $\hat{K}$: exchange operator with

$$E_{\text{x}}(\Phi_{\text{HF}}) = \frac{1}{2} \langle \Phi_{\text{HF}} | \hat{K} | \Phi_{\text{HF}} \rangle = -\frac{1}{2} \int \int \frac{|\gamma_{\text{HF}}(\mathbf{r}; \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}',$$

where γ_{HF} is the one-body reduced HF density matrix.

- $\hat{F}$: Fock operator $\hat{F} = \hat{T} + \hat{V} + \hat{J} + \hat{K}$ with

$$\Lambda_{\text{HF}}(\nu) = \langle \Phi_{\text{HF}} | \hat{F} | \Phi_{\text{HF}} \rangle \quad \text{and} \quad E_{\text{HF}}(\nu) = \Lambda_{\text{HF}}(\nu) - E_{\text{H}}[\rho_{\text{HF}}] - E_{\text{x}}(\Phi_{\text{HF}}).$$

MPAC Hamiltonian: $\hat{H}_{\text{MPAC}}^\lambda = \hat{T} + \hat{V} + \lambda \hat{V}_{\text{ee}} + (1 - \lambda)(\hat{J} + \hat{K})$

MPAC energy and density:

$$E_{\text{MPAC}}^\lambda(\nu) = \min_{\|\Psi\|=1} \langle \Psi | \hat{H}_{\text{MPAC}}^\lambda | \Psi \rangle = \langle \Psi^\lambda | \hat{H}_{\text{MPAC}}^\lambda | \Psi^\lambda \rangle \text{ and } \Psi^\lambda \rightarrow \rho^\lambda.$$

Physical system $\lambda = 1$

$\Psi^{\lambda=1} = \Psi_{\text{DFT}}, \rho^{\lambda=1} = \rho$ and

$$E_{\text{MPAC}}^{\lambda=1}(\nu) = E_0(\nu) = E_{\text{DFT}}[\rho]$$

HF reference system $\lambda = 0$

$\Psi^{\lambda=0} = \Phi_{\text{HF}}, \rho^{\lambda=0} = \rho_{\text{HF}}$ and

$$E_{\text{MPAC}}^{\lambda=0}(\nu) = E_{\text{HF}}(\nu) + E_{\text{H}}[\rho_{\text{HF}}] + E_{\text{x}}(\Phi_{\text{HF}})$$

MPAC Correlation Energy

$$E_{\text{c}}(\nu) = E_0(\nu) - E_{\text{HF}}(\nu) = \langle \Psi^{\lambda=1} | \hat{H}^{\lambda=1} | \Psi^{\lambda=1} \rangle - \langle \Phi_{\text{HF}} | \hat{H}^{\lambda=1} | \Phi_{\text{HF}} \rangle = \int_0^1 \mathcal{E}_{\text{c}}^\lambda d\lambda$$

$$\text{MPAC integrand: } \mathcal{E}_{\text{c}}^\lambda = \langle \Psi^\lambda | \hat{V}_{\text{ee}} - \hat{J} - \hat{K} | \Psi^\lambda \rangle - \langle \Phi_{\text{HF}} | \hat{V}_{\text{ee}} - \hat{J} - \hat{K} | \Phi_{\text{HF}} \rangle.$$

Small- λ expansion observed as MP perturbation series: $\mathcal{E}_{\text{c}}^{\lambda \rightarrow 0} = \sum_{n=2}^{\infty} n E_{\text{c}}^{\text{MP}n} \lambda^{n-1}$.

M. Seidl, S. Giarrusso, S. Vuckovic, E. Fabiano and P. Gori-Giorgi, *J. Chem. Phys.* **149**(24), 2018.

K. J. Daas, H. Zhao, **E.P.** and S. Vuckovic, *J. Chem. Theory Comput.* **21** (11), 5501–5513, 2025.

C. Møller and M. S. Plesset, *Phys. Rev.* **46** (7), 618–622, 1934.

Local Adiabatic Connection

Fix $v \in \mathcal{V} \Rightarrow E_c(v) = E_c$.

Correlation Energy Density

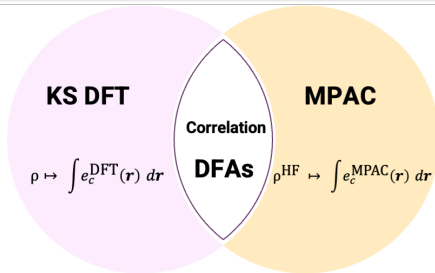
$$E_c = \int e_c(\mathbf{r}) d\mathbf{r} \text{ and } \mathcal{E}_c^\lambda = \int \varepsilon_c^\lambda(\mathbf{r}) d\mathbf{r} \Rightarrow e_c(\mathbf{r}) = \int_0^1 \varepsilon_c^\lambda(\mathbf{r}) d\lambda$$

MPAC: v -fixed (choose $v(\mathbf{r})$ to obtain MPAC energy densities)

DFT AC: ρ -fixed (choose $\rho(\mathbf{r})$ to obtain DFT AC energy densities)

General Gauge Strategy

- Electrostatic potential of the λ -dependent correlation hole
- Conventional gauge for Kohn-Sham (KS) density functional approximations (DFAs)
- Natural gauge in MPAC linked to traditional quantum-chemical E_c .



Local mapping of correlation

Hellmann-Feynman MPAC

Theorem (Integral MPAC Energy)

Let Ψ^λ be a non-degenerate eigenstate of $\hat{H}_{\text{MPAC}}^\lambda$ with isolated eigenvalue E_{MPAC}^λ for $\lambda \in \mathbb{R}_0^+$, then

$$E_{\text{MPAC}}^\lambda = E_{\text{MPAC}}^{\lambda=0} + \int_0^\lambda \langle \Psi^\mu | \hat{V}_{\text{ee}} - \hat{J} - \hat{K} | \Psi^\mu \rangle d\mu.$$

Proof:

$$\begin{aligned} E_{\text{MPAC}}^\lambda - E_{\text{MPAC}}^{\lambda=0} &= \int_0^\lambda \frac{d}{d\mu} E_{\text{MPAC}}^\mu d\mu \\ &= \int_0^\lambda \frac{d}{d\mu} \langle \Psi^\mu | \hat{H}_{\text{MPAC}}^\mu | \Psi^\mu \rangle d\mu \\ &\stackrel{\text{Hellmann-Feynman}}{=} \int_0^\lambda \left\langle \Psi^\mu \left| \frac{d}{d\mu} \hat{H}_{\text{MPAC}}^\mu \right| \Psi^\mu \right\rangle d\mu \\ &= \int_0^\lambda \langle \Psi^\mu | \hat{V}_{\text{ee}} - \hat{J} - \hat{K} | \Psi^\mu \rangle d\mu. \end{aligned}$$

□

Hellmann-Feynman MPAC

Corollary (Integral MPAC Correlation Energy)

$$E_c = \int_0^1 \mathcal{E}_c^\lambda d\lambda \text{ with } \mathcal{E}_c^\lambda = \langle \Psi^{\lambda=1} | \hat{V}_{ee} - \hat{J} - \hat{K} | \Psi^{\lambda=1} \rangle - \langle \Phi_{HF} | \hat{V}_{ee} - \hat{J} - \hat{K} | \Phi_{HF} \rangle.$$

Proof:

$$\begin{aligned} E_c &= \langle \Psi^{\lambda=1} | \hat{H}_{\text{MPAC}}^{\lambda=1} | \Psi^{\lambda=1} \rangle - \langle \Psi^{\lambda=0} | \hat{H}_{\text{MPAC}}^{\lambda=1} | \Psi^{\lambda=0} \rangle \\ &= E_{\text{MPAC}}^{\lambda=1} - \langle \Psi^{\lambda=0} | \hat{T} + \hat{V} + \hat{J} + \hat{K} + \hat{V}_{ee} - \hat{J} - \hat{K} | \Psi^{\lambda=0} \rangle \end{aligned}$$

□

Hellmann-Feynman MPAC

Corollary (Integral MPAC Correlation Energy)

$$E_c = \int_0^1 \mathcal{E}_c^\lambda d\lambda \text{ with } \mathcal{E}_c^\lambda = \langle \Psi^{\lambda=1} | \hat{V}_{ee} - \hat{J} - \hat{K} | \Psi^{\lambda=1} \rangle - \langle \Phi_{HF} | \hat{V}_{ee} - \hat{J} - \hat{K} | \Phi_{HF} \rangle.$$

Proof:

$$\begin{aligned} E_c &= \langle \Psi^{\lambda=1} | \hat{H}_{\text{MPAC}}^{\lambda=1} | \Psi^{\lambda=1} \rangle - \langle \Psi^{\lambda=0} | \hat{H}_{\text{MPAC}}^{\lambda=1} | \Psi^{\lambda=0} \rangle \\ &= E_{\text{MPAC}}^{\lambda=1} - \langle \Psi^{\lambda=0} | \hat{T} + \hat{V} + \hat{J} + \hat{K} + \hat{V}_{ee} - \hat{J} - \hat{K} | \Psi^{\lambda=0} \rangle \\ &= E_{\text{MPAC}}^{\lambda=1} - E_{\text{MPAC}}^{\lambda=0} - \langle \Psi^{\lambda=0} | \hat{V}_{ee} - \hat{J} - \hat{K} | \Psi^{\lambda=0} \rangle \\ &= \int_0^1 \langle \Psi^\lambda | \hat{V}_{ee} - \hat{J} - \hat{K} | \Psi^\lambda \rangle d\lambda - \langle \Phi_{HF} | \hat{V}_{ee} - \hat{J} - \hat{K} | \Phi_{HF} \rangle \end{aligned}$$

□

What about the degenerate case?

Convex Analytic AC Formalism

Mixed state representation via N -particle density matrix operators:

$$\mathcal{G}_N = \left\{ \Gamma : \mathcal{H}_N \rightarrow \mathcal{H}_N \mid \Gamma = \Gamma^\dagger, \Gamma \geq 0, \text{Tr } \Gamma = 1 \right\}$$

- Integral kernel form: $\Gamma : \mathcal{H}_N \rightarrow \Gamma(\mathbf{r}_1, \dots, \mathbf{r}_N; \mathbf{r}'_1, \dots, \mathbf{r}'_N)$.
- One-body reduced density matrix:

$$\gamma(\mathbf{r}; \mathbf{r}') = N \int \Gamma(\mathbf{r}, \mathbf{r}_2, \dots, \mathbf{r}_N; \mathbf{r}', \mathbf{r}_2, \dots, \mathbf{r}_N) d\mathbf{r}_2 \cdots d\mathbf{r}_N, \quad \gamma(\mathbf{r}; \mathbf{r}) = \rho(\mathbf{r}).$$

- For $\rho \in \mathcal{P}_N$ there exist at least one¹ $\Gamma \in \mathcal{G}_N$ such that $\Gamma \rightarrow \rho_\Gamma$.
- HF solution: $\Gamma_{\text{HF}} \rightarrow \gamma_{\text{HF}} \rightarrow \rho_{\text{HF}}$.

Lemma (Concavity of MPAC Energy)

The MPAC energy for mixed states $E_{\text{MPAC}}^\lambda = \inf_{\Gamma \in \mathcal{G}_N} \text{Tr} \left(\hat{H}_{\text{MPAC}}^\lambda \Gamma \right)$ is concave in $\lambda \in \mathbb{R}_0^+$.

Proof: Consider $\lambda_1, \lambda_2 \geq 0$ and $s \in [0, 1]$ for $\tilde{\lambda} = s\lambda_1 + (1-s)\lambda_2$:

$$E_{\text{MPAC}}^{\tilde{\lambda}} = \inf_{\Gamma \in \mathcal{G}_N} \text{Tr} \left(\hat{H}_{\text{MPAC}}^{\tilde{\lambda}} \Gamma \right) = \inf_{\Gamma \in \mathcal{G}} \left\{ s \text{Tr} \left(\hat{H}_{\text{MPAC}}^{\lambda_1} \Gamma \right) + (1-s) \text{Tr} \left(\hat{H}_{\text{MPAC}}^{\lambda_2} \Gamma \right) \right\} \geq s E_{\text{MPAC}}^{\lambda_1} + (1-s) E_{\text{MPAC}}^{\lambda_2}.$$

□

¹E. H. Lieb, *Int. J. Quantum Chem.* **24** (3), 243–277, 1983.

Integral AC Energy

Consider Γ^λ as the minimizer for $E_{\text{MPAC}}^\lambda = \inf_{\Gamma \in \mathcal{G}_N} \text{Tr} \left(\hat{H}_{\text{MPAC}}^\lambda \Gamma \right) \Rightarrow \Gamma^{\lambda=0} = \Gamma_{\text{HF}}$.

Theorem (Integral MPAC energy for mixed states)

For every $\lambda \in \mathbb{R}_0^+$, the MPAC energy is adiabatically connected as

$$E_{\text{MPAC}}^\lambda(v) = E_{\text{MPAC}}^{\lambda=0} + \int_0^\lambda \text{Tr} \left[\left(\hat{V}_{ee} - \hat{J} - \hat{K} \right) \Gamma^\mu \right] d\mu.$$

Corollary (Integral MPAC correlation energy for mixed states)

For every $\lambda \in \mathbb{R}^+$, the generalised MPAC correlation energy

$\lambda E_c^\lambda = E_{\text{MPAC}}^\lambda - E_{\text{HF}} - (1 - \lambda) (E_{\text{H}}[\rho_{\text{HF}}] + E_{\text{x}}(\Gamma_{\text{HF}}))$ is adiabatically connected as

$$\lambda E_c^\lambda = \int_0^\lambda \mathcal{E}_c^\mu d\mu = \int_0^\lambda \text{Tr} \left[\left(\hat{V}_{ee} - \hat{J} - \hat{K} \right) (\Gamma^\mu - \Gamma_{\text{HF}}) \right] d\mu.$$

Proof:

- $\lambda = 1$: $E_c^{\lambda=1} = E_{\text{MPAC}}^{\lambda=1} - E_{\text{HF}} = E_c$ and $E_c = \int_0^1 \mathcal{E}_c^\mu d\mu$.
- $\lambda \in \mathbb{R}^+$: $E_{\text{HF}} = E_{\text{MPAC}}^{\lambda=0} - E_{\text{H}}[\rho_{\text{HF}}] - E_{\text{x}}(\Gamma_{\text{HF}})$ and use integral MPAC energy for mixed states (similar to previous derivation of E_c).

□

Proof of Theorem:

- Concavity of E_{MPAC}^λ in $\lambda \in \mathbb{R}_0^+$ and linearity of the trace

$$\Rightarrow \text{Tr} \left[\left(\hat{V}_{\text{ee}} - \hat{J} - \hat{K} \right) \Gamma^\lambda \right] \in \bar{\partial}_\lambda E_{\text{MPAC}}^\lambda.$$

- *Superdifferential* of a concave and upper semicontinuous function $f : \mathbb{R} \rightarrow \mathbb{R}$:

$$\bar{\partial}_x f(x) = \{f \in \mathbb{R} \mid \forall y \in \mathbb{R} : f(x) + (y - x) \cdot f \geq f(y)\}$$

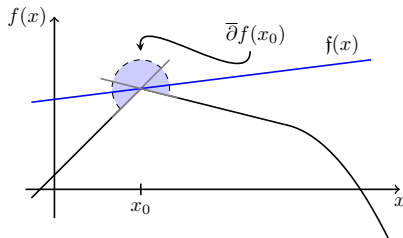


Figure: Superdifferential of a concave functional at x_0 as the shaded area.
 $f(x)$ is the slope of one element in $\bar{\partial}f(x_0)$.

- Generalised Newton-Leibnitz formula for any $\mathfrak{E}^\lambda \in \bar{\partial}_\lambda E_{\text{MPAC}}^\lambda$:

$$E_{\text{MPAC}}^\lambda - E_{\text{MPAC}}^{\lambda=0} = \int_0^\lambda \mathfrak{E}^\mu d\mu.$$

□

Example

$$\begin{aligned}\text{MPAC Energy: } E_{\text{MPAC}}^{\lambda} &= E_{\text{MPAC}}^{\lambda=0} + \int_0^{\lambda} \text{Tr} \left[\left(\hat{V}_{\text{ee}} - \hat{J} - \hat{K} \right) \Gamma^{\lambda} \right] d\mu \\ &= E_{\text{HF}} + (1 - \lambda) (E_{\text{H}}[\rho_{\text{HF}}] + E_{\text{x}}(\Gamma_{\text{HF}})) + \lambda E_{\text{c}}^{\lambda}\end{aligned}$$

Numerical procedure:

1. Solve HF system and obtain $E_{\text{MPAC}}^{\lambda=0} = E_{\text{HF}} + E_{\text{H}}[\rho_{\text{HF}}] + E_{\text{x}}(\Gamma_{\text{HF}})$.
2. Obtain Γ^{λ} (minimizer of $E_{\text{MPAC}}^{\lambda}$) at every $\lambda \in [0, 1]$ (e.g. full configuration interaction (FCI)).
3. Perform integration over a discretized λ -grid.

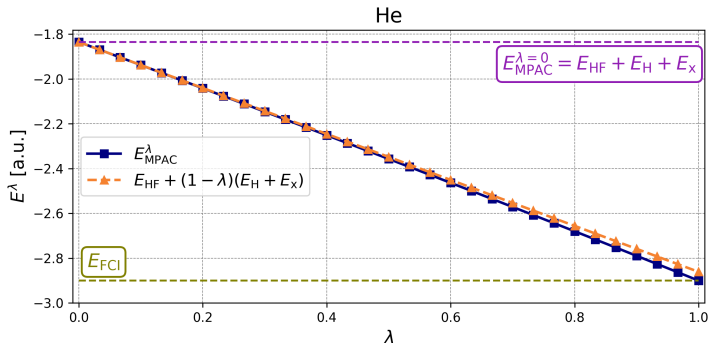


Figure: MPAC energy with respect to λ .

Properties of the AC Integrand

MPAC integrand monotonicity

For Γ^λ as the minimizer of E_{MPAC}^λ , the MPAC integrand

$$\mathcal{E}_c^\lambda = \text{Tr} \left[\left(\hat{V}_{\text{ee}} - \hat{J} - \hat{K} \right) \left(\Gamma^\lambda - \Gamma_{\text{HF}} \right) \right]$$

is decreasing in $\lambda \in \mathbb{R}^+$ and $\lambda \mathcal{E}_c^\lambda \leq \lambda E_c^\lambda \leq 0$.

DFT AC integrand monotonicity¹

For $\Gamma_{\text{DFT}}^\lambda$ as the minimizer of E_{DFT}^λ , the DFT AC integrand

$$\mathcal{E}_{\text{c,DFT}}^\lambda[\rho] = \text{Tr} \left(\hat{V}_{\text{ee}} \left[\Gamma_{\text{DFT}}^\lambda - \Gamma_{\text{DFT}}^{\lambda=0} \right] \right)$$

is decreasing in $\lambda \in \mathbb{R}^+$ and $E_{\text{c,DFT}}^\lambda \leq 0$.

¹A. M. Teale, S. Coriani and T. Helgaker, *J. Chem. Phys.* **132**, 164115, 2010.

MPAC correlation energy at zero-coupling limit

$$\lim_{\lambda \rightarrow 0^+} E_c^\lambda = \lim_{\lambda \rightarrow 0^+} \frac{1}{\lambda} \int_0^\lambda \mathcal{E}_c^\mu d\mu = \lim_{\lambda \rightarrow 0^+} \frac{1}{\lambda} \int_0^\lambda \text{Tr} \left[\left(\hat{V}_{ee} - \hat{J} - \hat{K} \right) (\Gamma^\mu - \Gamma_{\text{HF}}) \right] d\mu \stackrel{?}{=} 0.$$

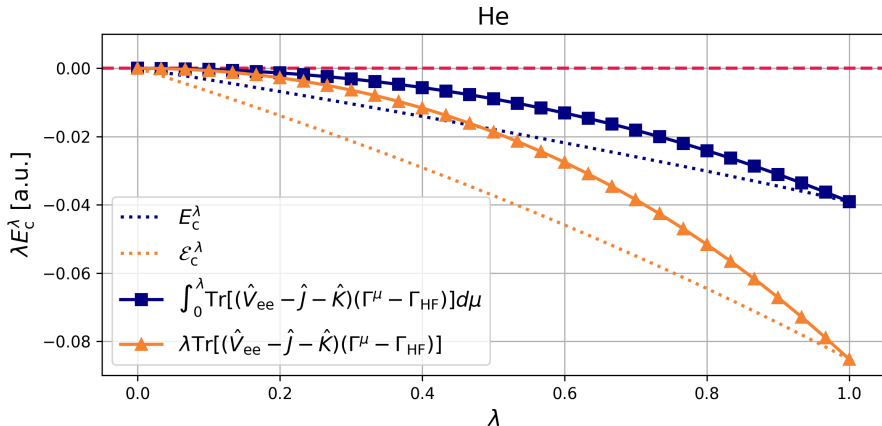


Figure: MPAC correlation energy with respect to λ .

A Virial Relation

Theorem (MPAC Virial Relation)

For every $\lambda \in \mathbb{R}^+$ and assuming $\mathcal{V} \subset H^1$, the virial relation for mixed states in MPAC is

$$\text{Tr} \left[\left(2\hat{T} + \lambda \hat{V}_{ee} + (1 - \lambda)(\hat{J} + \hat{K}) \right) \Gamma^\lambda \right] = \int \rho_{\Gamma^\lambda}(\mathbf{r}) \mathbf{r} \cdot \nabla v(\mathbf{r}) d\mathbf{r}.$$

Proof:

- Applying Ehrenfest theorem for mixed states with linear operator $\hat{A} = \sum_{j=1}^N \mathbf{r}_j \cdot \nabla_{\mathbf{r}_j}$.
- Taking the trace against Γ^λ . □

DFT AC Virial Relation

Let $\Gamma_{\text{DFT}}^\lambda$ be the minimizer of $E_{\text{DFT}}^\lambda[\rho]$ (i. e. v^λ such that $\Gamma_{\text{DFT}}^\lambda \rightarrow \rho$), then

$$\text{Tr} \left[\left(2\hat{T} + \lambda \hat{V}_{ee} \right) \Gamma_{\text{DFT}}^\lambda \right] = \int \rho(\mathbf{r}) \mathbf{r} \cdot \nabla v^\lambda(\mathbf{r}) d\mathbf{r}.$$

I. Theophilou, M. Penz, M. Ruggenthaler, and A. Rubio, "Virial relations for electrons coupled to quantum field models", *J. Chem. Theory Comput.* **16**, 6236 - 6243, 2020.

A. Laestadius, M. A. Csirik, M. Penz, N. Tancogne-Dejean, M. Ruggenthaler, A. Rubio, and T. Helgaker, "Exchange-only virial relation from the adiabatic connection", *J. Chem. Phys.* **160**, 2024.

Exchange Only Virial Relation

DFT AC

Exchange only DFT AC virial relation:

$$E_x[\rho] = E_H[\rho] - \lim_{\lambda \rightarrow 0^+} \int \rho(\mathbf{r}) \mathbf{r} \cdot \nabla v^\lambda(\mathbf{r}),$$

MPAC Conjecture

Exchange only MPAC virial relation:

$$E_x(\Gamma_{\text{HF}}) = E_H[\rho_{\text{HF}}] - \lim_{\lambda \rightarrow 0^+} \frac{1}{\lambda} \int (\rho_{\Gamma^\lambda}(\mathbf{r}) - \rho_{\text{HF}}(\mathbf{r})) \mathbf{r} \cdot \nabla v(\mathbf{r}) d\mathbf{r}.$$

I. Theophilou, M. Penz, M. Ruggenthaler, and A. Rubio, "Virial relations for electrons coupled to quantum field models", *J. Chem. Theory Comput.* **16**, 6236 - 6243, 2020.

A. Laestadius, M. A. Csirik, M. Penz, N. Tancogne-Dejean, M. Ruggenthaler, A. Rubio, and T. Helgaker, "Exchange-only virial relation from the adiabatic connection", *J. Chem. Phys.* **160**, 2024.

Conclusion and Outlook

- DFT AC is ρ -fixed: requires $v^\lambda \in \mathcal{V}$ such that $\Gamma_{\text{DFT}}^\lambda \rightarrow \rho$ to obtain E_{DFT}^λ
- MPAC is v -fixed: fix $v \in \mathcal{V}$ to obtain E_{MPAC}^λ and ρ_{Γ^λ}
- Convex analytic AC formalism for mixed states
 - Integral MPAC relations
 - MPAC integrand properties
 - MPAC virial relation
- Numerical investigation for (near) degenerate systems
- Explore $\lambda > 1$ case
- Exchange only virial relation in MPAC

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Thank you for your attention!